Final Report

# The Art of Scientific Computation: Spectral and Mixing Properties of Supersingular Isogeny Graphs

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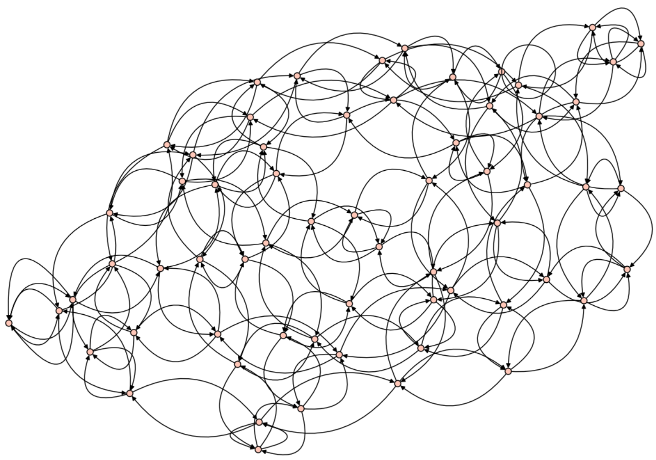
COMP90072



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# Chapter 1

# Introduction

## 1.1 Overview

In this project, I will construct and analyse supersingular $\ell$-isogeny graphs over finite fields $\mathbb{F}_{p^2}$. These are graphs whose vertices represent isomorphism classes of supersingular elliptic curves over $\mathbb{F}_{p^2}$ and whose edges represent degree-$\ell$ isogenies between them (see [5] for background). These graphs are of special interest in cryptography, since many cryptographic protocols can be understood as applying random walks on these graphs. One such example is given in [1], whose designs provide inspiration for the graphs considered in this project.

The significance of these graphs in cryptography arises mainly from their strong expansion properties; in particular, they are Ramanujan graphs, and therefore random walks on these graphs converge quickly to a uniform distribution. The most cryptographically interesting consequence is that given any two vertices in one such graph, it will be computationally expensive to find a path connecting them.

We are therefore interested in performing empirical verification of theoretical properties attributed to these graphs. To this end, two main interests stand out: the eigenvalue spectra associated to the adjacency matrices of these graphs, and the mixing times of random walks on these graphs. For this project, I will produce a sample of supersingular isogeny graphs over a range of parameters and compute these two characteristics, comparing them to theoretical estimates in order to illustrate and corroborate them.

## 1.2 Scope

We have the following primary objectives.

1. Efficiently compute supersingular $\ell$-isogeny graphs over $\mathbb{F}_{p^2}$ for a range of parameters $p$ and $\ell$.

2. Compute the adjacency matrices for a sample of supersingular isogeny graphs and compute their eigenvalue spectra. We will use this to answer two questions.

   (a) How does the spectral gap of a supersingular $\ell$-isogeny graph over $\mathbb{F}_{p^2}$ vary according to the parameters $\ell$ and $p$?

   (b) How does the spectral gap of a supersingular $\ell$-isogeny graph over $\mathbb{F}_{p^2}$, as measured empirically, align with theoretical estimates of the same quantity?

3. Produce, for each sampled supersingular $\ell$-isogeny graph over $\mathbb{F}_{p^2}$, a sample of random walks on that graph. We will use this to answer two questions.

   (a) How are mixing times of random walks distributed for each graph, in terms of parameters $\ell$ and $p$?

   (b) How do the mixing times of random walks on these graphs, as measured empirically, align with theoretical estimates of mixing times for Ramanujan graphs?

# Chapter 2

# Mathematical Background

This chapter will cover some of the relevant concepts and vocabulary for this project, as well as an outline of the properties we will wish to assess. While some background in group theory is assumed, we aim to remain light on technicalities, focusing mainly on the key terms required to understand the problem at hand. The interested reader may consult [5], [2], or [4] for further information; indeed, the following can be treated as an abbreviated summary of these sources. The sections on spectral graph theory, and on random walks and mixing times, are drawn from [3].

## 2.1 Elliptic curves

The definition of elliptic curve can be rendered in various equivalent forms. For instance, as per [6], we could define an elliptic curve $E$ over a field $K$ as a non-singular projective algebraic curve of genus 1 having a distinguished $K$-rational point $O \in E$. But, we are interested in elliptic curves with a view to computation, and therefore give a simpler definition, more tractable for our purposes.

**Definition 1** (Elliptic Curve)**.** Let $K$ be a field of characteristic other than 2 or 3. Then an elliptic curve $E$ over $K$ consists of the set of solutions $(x, y) \in K^2$ of the short Weierstrass equation

$$E : y^2 = x^3 + ax + b$$

where $a, b \in K$ and $4a^3 + 27b^2 \neq 0$.

Over the real numbers $\mathbb{R}$, elliptic curves can be viewed as familiar planar curves.

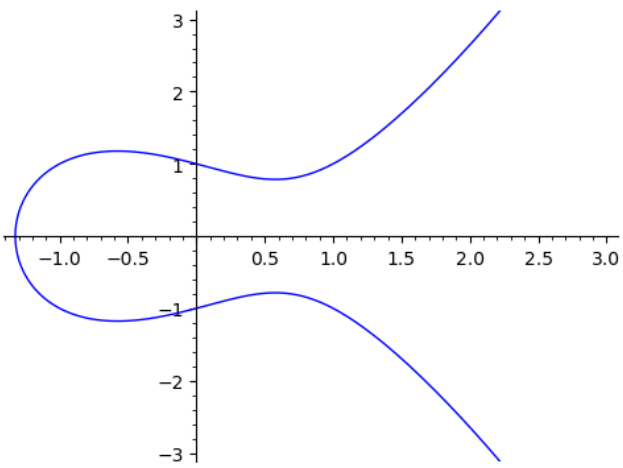


**Figure 2.1:** Elliptic curve $E : y^2 = x^3 - x + 1$ over $\mathbb{R}$.

However, we will be mainly interested in elliptic curves over finite fields $\mathbb{F}_{p^n}$ for prime $p$ and $n \in \mathbb{Z}_{\geq 1}$. An example of the solution set to one such curve over a finite field is shown below.

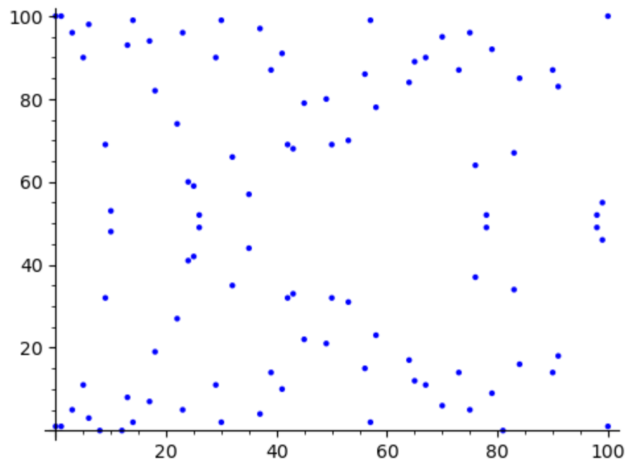


**Figure 2.2:** Elliptic curve $E : y^2 = x^3 - x + 1$ over $\mathbb{F}_{101}$, the finite field of order 101.

Key to the theory of elliptic curves is the fact that it is possible to define a group operation (conventionally called a *group law*) on the points of $E$, satisfying certain desirable properties. The definition of this operation arises from projective geometry, whose details we defer to a reference such as [6]. For our purposes, it will suffice to understand that any elliptic curve has a group structure.

**Definition 2** (Group Law)**.** Let $E$ be an elliptic curve over a field $K$. It is a consequence of Bézout's theorem that under mild assumptions and counting multiplicity, an algebraic curve of degree $m$ and an algebraic curve of degree $n$ will intersect in $mn$ points. Since an elliptic curve has degree 3 and a line has degree 1, it follows that the line spanned by two points $P, Q \in E$ will intersect $E$ at a unique third point $R' \in E$. We let $R \in E$ be the reflection of $R'$ over the $x$-axis and define a *group law* $P \oplus Q = R$, such that $(E, \oplus)$ forms an abelian group. (For details, see [6].)

## 2.2 Isogenies

If $E_1$ and $E_2$ are elliptic curves over a common field $K$, then a morphism of elliptic curves is a rational map $f\colon E_1 \to E_2$. Loosely, this can be understood as a map defined by rational functions. However, we will be interested in a more restrictive notion of mapping between elliptic curves, namely *isogeny*.

**Definition 3** (Isogeny)**.** Let $E_1$ and $E_2$ be elliptic curves over a common field $K$. An *isogeny* of elliptic curves is a morphism $f\colon E_1 \to E_2$ which is also a surjective group homomorphism with respect to the group structures on $E_1$ and $E_2$.

Isogenies provide a notion of mapping between elliptic curves which respects both the geometric and arithmetic structures of the curve, and which can therefore encode useful information about the curves thereby related. We will enumerate some useful properties.

**Definition 4** (Elliptic curve isomorphism)**.** An isogeny $f\colon E_1 \to E_2$ is called an *isomorphism* if it is an isomorphism of the underlying groups in $E_1$ and $E_2$.

**Definition 5** (Degree of an isogeny)**.** Suppose an isogeny $f\colon E_1 \to E_2$ is separable*. Then $f$ has a quantity called the *degree*, denoted by $\deg(f)$, which can be computed by $\deg(f) = |\ker(f)|$.

In fact, over suitable fields, the kernel of an isogeny will uniquely determine that isogeny.

We next consider the possibility of defining an isogeny between a curve and itself.

**Definition 6** (The endomorphism ring)**.** Let $E$ be an elliptic curve over a field $K$. An isogeny $f\colon E \to E$ is called an endomorphism of $E$. The set $\mathrm{End}(E)$ of all endomorphisms of $E$, including also the zero morphism $0\colon E \to E$, forms a ring where addition is defined pointwise and multiplication is defined to be the composition operation. That is, $fg = f \circ g$ and for $f, g \in \mathrm{End}(E)$ we define $f + g$ to be the isogeny $x \mapsto f(x) + g(x)$. The resulting ring $\mathrm{End}(E)$ is called the *endomorphism ring of $E$*.

*For our purposes, all isogenies will be separable. The reader can therefore, if so desired, treat this condition as a technicality.

Since any elliptic curve $E$ forms an abelian group, we can define integer multiplication of points on the curve using the additive group structure. Then, it holds for any elliptic curve $E$ that the multiplication-by-$n$ map $x \mapsto nx$ is an endomorphism $[n]\colon E \to E$. So, for any elliptic curve $E$, $\mathrm{End}(E)$ has a subring isomorphic to $\mathbb{Z}$.

## 2.3 Ordinary and supersingular

Elliptic curves over finite fields can be distinguished as either *ordinary* or *supersingular*. Ordinary elliptic curves form well understood structures; we will be interested in the alternate, supersingular case.

**Definition 7.** Let $E$ be an elliptic curve over a finite field $K$. If $\mathrm{End}(E)$ is isomorphic to an order in an imaginary quadratic field $\mathbb{Q}(\sqrt{-d})$, we say that $E$ is *ordinary*. In this case, $\mathrm{End}(E)$ is commutative.

If instead $\mathrm{End}(E)$ is isomorphic to an order in a quaternion algebra, we say that $E$ is *supersingular*. In this case, $\mathrm{End}(E)$ is non-commutative.

All elliptic curves over finite fields are either ordinary or supersingular, and we see that supersingular curves can be distinguished by having larger endomorphism rings. This can be understood, morally, as indicating that supersingular elliptic curves have more complex internal structure.

**Property 1.** *Over any field of characteristic $p$, if $E$ is a supersingular elliptic curve then the $j$-invariant $j(E)$ is defined over $\mathbb{F}_{p^2}$. And, there are $N_p = \lfloor \frac{p}{12} \rfloor + \varepsilon_p$ such supersingular elliptic curves over characteristic $p$, where $\varepsilon_p = 0, 1, 1, 2$ corresponding to the cases that $p \equiv 1, 5, 7, 11 \pmod{12}$ respectively.*

Supersingular elliptic curves are named so because they are singular in the sense of rarity; over any field, few supersingular elliptic curves exist compared to ordinary elliptic curves.

## 2.4 The $j$-invariant

Isomorphisms are typically regarded as providing a notion of *equivalence*; if two objects are isomorphic within some category, then their properties within that category must be identical. So, we are often interested in classifying objects *up to isomorphism.* For elliptic curves, isomorphism classes of elliptic curves have a convenient classification by the *$j$-invariant.* This invariant has a more general definition, but we give a specific form applicable to curves expressed by Weiestrass equations.

**Definition 8** ($j$-invariant)**.** Let $E$ be an elliptic curve over a field $K$, and assume that $E$ is given by the Weierstrass equation $y^2 = x^3 + ax + b$. Then the *$j$-invariant* of $E$ is

$$j(E) = 1728 \frac{4a^3}{4a^3 + 27b^2}.$$

When the base field $K$ is algebraically closed, such as in the case that $K = \mathbb{C}$, then two elliptic curves $E_1, E_2$ over $K$ are isomorphic if and only if $j(E_1) = j(E_2)$. So, in this case, the $j$-invariant classifies all elliptic curves up to isomorphism.

Later we will consider elliptic curves over $\mathbb{F}_{p^2}$, where $p$ is prime. In this case, although $\mathbb{F}_{p^2}$ is not algebraically closed, every supersingular elliptic curve has its $j$-invariant defined as an element of $\mathbb{F}_{p^2}$, and therefore the $j$-invariant can still be used as a classification of isomorphism classes in the algebraic closure of $\mathbb{F}_{p^2}$.

## 2.5 Supersingular $\ell$-isogeny graphs

An *isogeny graph* is, in general, a graph whose vertices correspond to isomorphism classes of elliptic curves, and whose edges are isogenies between those (isomorphism classes of) elliptic curves. We are interested in certain *supersingular* isogeny graphs, whose edges are restricted to isogenies of common fixed degree.

**Definition 9** ($\ell$-isogeny graphs)**.** Let $K$ be a field. For $\ell \in \mathbb{Z}_{\geq 0}$, the $\ell$-isogeny graph over $K$ is the graph $G_\ell(K)$ whose vertices are isomorphism classes of elliptic curves over $K$, and whose edges are $\ell$-isogenies (i.e. isogenies of degree $\ell$).

We will hereafter restrict our attention to the case that $K = \mathbb{F}_{p^2}$, where $p \in \mathbb{Z}$ is prime. In this case, the supersingular elliptic curves over $\mathbb{F}_{p^2}$ form a connected component within $G_\ell(\mathbb{F}_{p^2})$, having approximately $p/12$ vertices. As we will see, this graph has many desirable properties.

**Definition 10** (Supersingular $\ell$-isogeny graphs)**.** Let $\ell, p \in \mathbb{Z}$ be primes such that $\ell \neq p$. The set of supersingular elliptic curves over $\mathbb{F}_{p^2}$ forms a connected component within $G_\ell(\mathbb{F}_{p^2})$, and is called the *supersingular $\ell$-isogeny graph over* $\mathbb{F}_{p^2}$, or alternatively the *supersingular component of* $G_\ell(\mathbb{F}_{p^2})$.

**Property 2.** *Any supersingular $\ell$-isogeny graph over $\mathbb{F}_{p^2}$ is finite, connected and $(\ell+1)$-regular.*

## 2.6 Spectral graph theory

Spectral graph theory pursues the insight that the various matrices associated with a graph, such as the adjacency matrix, encode in their eigenvalue spectra useful information pertaining to their underlying graphs.

**Definition 11** (Adjacency matrix)**.** Let $G$ be a graph with vertex set $V = \{v_1, \dots, v_n\}$ and edge set $E$. The adjacency matrix of $G$ is the $n \times n$ matrix $A$ whose $(i, j)$-th entry $a_{ij}$ is equal to the number of edges in $E$ beginning at $v_i$ and ending at $v_j$.

It is a simple result of linear algebra that if the adjacency matrix $A$ of a graph is symmetric, then its eigenvalues are all real. In particular, this will hold for either undirected graphs, or for symmetric directed graphs.

**Definition 12** (Ramanujan graph)**.** Suppose $G$ is a connected, $k$-regular graph with $n$ vertices and adjacency matrix $A$. Suppose the eigenvalues $\lambda_1, \dots, \lambda_n$ are sorted by decreasing absolute value,

$$|\lambda_1| \geq |\lambda_2| \geq \cdots \geq |\lambda_{n-1}| \geq |\lambda_n|.$$

Then $G$ is a *Ramanujan graph* if

$$|\lambda_2| \leq 2\sqrt{k-1}.$$

In particular, in the case that $G$ has symmetric adjacency matrix $A$, the eigenvalues of $A$ satisfy

$$k = \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_{n-1} \geq \lambda_n \geq -k.$$

Now, recalling that the supersingular $\ell$-isogeny graph over $\mathbb{F}_{p^2}$ is connected and $(\ell+1)$-regular, we have the following property.

**Property 3.** *The supersingular $\ell$-isogeny graph over $\mathbb{F}_{p^2}$ is a Ramanujan graph, satisfying*

$$|\lambda_2| \leq 2\sqrt{\ell}.$$

## 2.7 Random walks and mixing times

We have seen that supersingular $\ell$-isogeny graphs over $\mathbb{F}_{p^2}$ are Ramanujan graphs. The most important consequence of this property is that random walks on these graphs have optimal mixing times.

**Definition 13** (Random walk)**.** Let $G$ be a graph. Supposing a vertex $v$ of $G$ has at least one outgoing edge, a *random step* on this graph starting $v$ is a uniformly random choice from the set of neighbours of $v$ (possibly including $v$ itself, in the case that $G$ contains self-loops). A *random walk* on $G$ is a finite or countable sequence of random steps.

It follows from the fact that supersingular $\ell$-isogeny graphs are Ramanujan graphs that they satisfy the following essential property (see [3, Theorem 1] for proof).

**Property 4** (Random walk theorem)**.** *Let $j_0$ be the $j$-invariant corresponding to a supersingular elliptic curve in $\mathbb{F}_{p^2}$ and let $j$ be the $j$-invariant corresponding to the final point reached after a random walk of $n$ steps in the supersingular $\ell$-isogeny graph. Then for every $j' \in \mathbb{F}_{p^2}$ we have*

$$\left|\Pr(j = j') - \frac{1}{N_p}\right| \leq \left(\frac{2\sqrt{\ell}}{\ell + 1}\right)^n.$$

Indeed, the stationary distribution for random walks on any $k$-regular graph converges to the uniform distributon, and the above theorem guarantees that for supersingular $\ell$-isogeny graphs, the rate of this convergence is exponential, increasing with greater $\ell$.

In light of this, let us consider a random walk on a supersingular $\ell$-isogeny graph beginning at $j$-invariant $j_0$. Defining $P_k \in \mathbb{R}^{N_p}$ to be the row vector with a 1 in the $k$-th entry and zeroes in every other entry, and defining

$$P_n(i) = \Pr(\text{walk is at } j_i \text{ after } n \text{ steps}),$$

we find that the vectors $P_n$ for $n \geq 0$ describe the probability distributions for the destinations of random walks on this graph after $n$ steps.

Now, it is a standard result of the theory of discrete-time Markov chains that our random walk probability distribution evolves according to the rule $P_{n+1} = P_n M$ where $M = \frac{1}{1+\ell}A$, given that $A$ is the adjacency matrix of our supersingular $\ell$-isogeny graph. In particular, it follows that $P_n = P_0 M^n$.

Our claim is that $P_n$ converges to the following uniform distribution in a suitable sense.

**Definition 14** (Uniform distribution)**.** Recall that a supersingular $\ell$-isogeny graph has $N_p$ vertices. So, for any $j$-invariant $j_i$ in this graph, the uniform distribution assigns to $j_i$ the probability $U(i) = \frac{1}{N_p}$.

Our preferred measure of convergence will be the following.

**Definition 15** (Total variation distance)**.** The *total variation distance* between the probability distributions $P_n$ and $U$ is defined by

$$d_{TV} = \frac{1}{2}\sum_{i=1}^{N_p}\left|P_n(i) - \frac{1}{N_p}\right|.$$

This can be quantified by the following *mixing time*, which we will later numerically analyse.

**Definition 16** (Mixing time)**.** For $\varepsilon > 0$, the *mixing time* $t_{\text{mix}}(\varepsilon)$ quantifies convergence of $P_n$ to $U$ using

$$t_{\text{mix}}(\varepsilon) = \min\{n \geq 0 \mid d_{\text{TV}}(P_n, U) \leq \varepsilon\}.$$

# Chapter 3

# Computation

## 3.1 Elliptic curves in Sage

Sage is a software package built atop Python which is widely used in mathematical research for algebraic computation. Its facilities include:

- representing finite fields as Python objects;
- representing elliptic curves as Python objects;
- testing elliptic curves for supersingularity;
- computing isogenies between elliptic curves;
- computing $j$-invariants corresponding to elliptic curves;
- graph structures;

and more, which will be relevant to our use case. In our project, we will use Sage to handle heavier algebraic operations so that we can focus on the relevant computational research questions.

## 3.2 Generating supersingular $\ell$-isogeny graphs

In constructing supersingular $\ell$-isogeny graphs, we would like vertices to represent isomorphism classes of supersingular elliptic curves over $\mathbb{F}_{p^2}$. But, we have seen that (up to certain technical details), each such isomorphism class is uniquely specified by a $j$-invariant, taking a value in $\mathbb{F}_{p^2}$. So, rather than using any concrete representation of an elliptic curve, we can perform all our required elliptic curve operations in terms of values in $\mathbb{F}_{p^2}$, namely $j$-invariants. Indeed, given a value `j_val` in a Sage finite field $\mathbb{F}_{p^2}$, one obtains a Boolean

```
EllipticCurve(j=j_val).is_supersingular()
```

corresponding to whether a supersingular elliptic curve over $\mathbb{F}_{p^2}$ exists having $j$-invariant equal to `j_val`. So, $j$-invariants will provide vertex representations for our isogeny graphs.

Now, since the supersingular $\ell$-isogeny graph is known to be connected, it will suffice to first identify one supersingular $j$-invariant over $\mathbb{F}_{p^2}$, and then to perform a graph traversal to compute the rest of the graph. In our implementation, we have chosen to use a breadth-first search.

It still remains to understand, however, how we will compute the isogeny-neighbours of any such $j$-invariant. That is, given a $j$-invariant $j_0$ known to lie in our graph, how will we find the set of $j$-invariants $j_i$ for $i = 1, \ldots, \ell + 1$ such that there exists an isogeny $j_0 \to j_i$? There are two main options before us.

First, recall that in suitable ('separable') cases it is possible uniquely determine an isogeny by its kernel. And, since any isogeny $f\colon E_1 \to E_2$ is a group homomorphism, this kernel is a subgroup of $E_1$; it follows that an isogeny can be identified by finding the generators of its kernel. Supposing this can be done, one may use *Vélu's formulas* to compute explicitly the elliptic curve $E_2$ which is the image of an isogeny $f$ having the specified kernel. And, indeed, the facilities for this are provided in Sage. But, this will typically be more computationally expensive than we desire, especially since an initial sampling process is required in order to find generators for suitable subgroups in $E_1$.

So, we will instead use an algebraic device known as the *modular polynomial*. This is an integer valued polynomial in two variables $\Phi_\ell \in \mathbb{Z}[X, Y]$ such that $(X, Y)$ is a root of $\Phi$ if and only if there exists an $\ell$-isogeny between elliptic curves $E_1 \to E_2$ such that $j(E_1) = X$ and $j(E_2) = Y$. In the case that we consider $j$-invariants over finite fields such as $\mathbb{F}_{p^2}$, the same result is obtained by evaluating $\Phi_\ell(X, Y)$ over that field instead. So, suppose we know at least one $j$-invariant, say $j_0$, lying in our graph. In order to compute its $\ell$-isogeny neighbours then, it will suffice to find the roots of the polynomial $\Phi_\ell(j_0, Y)$. Indeed, in sage one can construct the modular polynomial by writing the following.

```
Phi = classical_modular_polynomial(Integer(l))
```

Then, for any known $j$-invariant `j1` lying in our graph, we can construct a polynomial

```
poly_y = Phi(j1, Y)
```

and find the neighbouring $j$-invariants of `j1` by computing the roots of `poly_y`, enumerating as follows.

```
for j2, mult in poly_y.roots():
    . . .
```

This will be our instrument for finding the neighbours of any vertex in our isogeny graph.

Now that we have established representations for the vertices in our graph, as well as for computing the neighbours of any vertex in our graph, there remain two computational steps. The first is to identify at least one initial supersingular $j$-invariant over $\mathbb{F}_{p^2}$, and the second is to perform a graph traversal beginning at that initial $j$-invariant. But, since $\mathbb{F}_{p^2}$ contains precisely $p^2$ elements, for moderate $p$ it will be possible simply to enumerate the elements of $\mathbb{F}_{p^2}$, and check each element for supersingularity as a $j$-invariant. Once an example is found, we can perform a breadth-first search using the modular polynomial to find neighbouring vertices in order to produce the entire supersingular $\ell$-isogeny graph.

## 3.3 An aside on exceptional $j$-invariants

We earlier claimed that for any prime $p$ and any prime $\ell \neq p$, the supersingular $\ell$-isogeny graph is $(\ell + 1)$-regular. This is true with a minor caveat: for some primes $p$, the elliptic curves having $j$-invariants 0 or 1728 may be supersingular, and may therefore be elements of our graph. But, these two (isomorphism classes of) elliptic curves are known to be exceptional, being the only two elliptic curves possessing what are known as 'extra automorphisms'. Consequently, if these are elements in our graph, the graph will fail to be $(\ell + 1)$-regular at these two points. Computationally this is minimally consequential, but for simplicity we will prefer to avoid this case altogether.

Now, it is known that the $j$-invariant 0 is supersingular over $\mathbb{F}_{p^2}$ if and only if $p \equiv 2 \pmod 3$, and that the $j$-invariant 1728 is supersingular over $\mathbb{F}_{p^2}$ if and only if $p \equiv 3 \pmod 4$. So, to avoid these cases, we will perform our experiments specialising to the cases that $p$ is a prime with $p \equiv 1 \pmod{12}$. To do this, we will write a function `first_n_suitable_primes` which returns a list of the first $n$ primes which are congruent to 1 modulo 12, beginning with 13. This function will be computed by simple enumeration, using primality checking provided by Sage.

## 3.4 Computing eigenvalue spectra

# Chapter 4

# Results

Discuss the results of computational experiments, produce plots of results against varying parameters $p$ and $\ell$. Produce visualisations of small isogeny graphs.

## Chapter 5

# Conclusion

Summarise findings, suggest further possible questions.

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